

423. Maximum internal assessment score = $\left(\frac{47}{50} \times 10\right) = 9.4$.

Minimum internal assessment score = $\left(\frac{14}{50} \times 10\right) = 2.8$.

$\therefore$ Required difference = $(9.4 - 2.8) = 6.6$.

424. Let the number of questions answered correctly be x .

Then, number of wrong answers = $(200 - x)$.

$\therefore 4x - (200 - x) = 200 \Rightarrow 5x = 400 \Rightarrow x = 80$.

425. Let the number of correct answers be x . Then, number of wrong answers = $(26 - x)$.

$\therefore 8x - 5(26 - x) = 0 \Rightarrow 13x = 130 \Rightarrow x = 10$.

426. Suppose the student attempted x questions of 2 marks correctly.

Then, number of 4 marks questions attempted correctly = $(15 - x)$.

$\therefore 2x + 4(15 - x) = 40 \Leftrightarrow 2x + 60 - 4x = 40$

$\Leftrightarrow 2x = 20 \Leftrightarrow x = 10$.

427. Let the number of correct answers be x . Then, number of incorrect answers = $(30 - x)$.

$\therefore x - 0.25(30 - x) = 13.75$

$\Leftrightarrow x - 7.5 + 0.25x = 13.75$

$\Leftrightarrow 1.25x = 21.25 \Leftrightarrow x = \frac{2125}{125} = 17$.

So, number of incorrect answers = $30 - 17 = 13$.

428. Suppose he hits the target x times.

Then, he misses it $(100 - x)$ times.

$\therefore x - (100 - x) = 30 \Rightarrow 2x = 130 \Rightarrow x = 65$.

So, number of times he misses the target = $100 - 65 = 35$.

429. Let the number of cows be x . Then, number of bulls = $2x$.

Number of caretakers = $\frac{x + 2x + 45}{15} = \frac{3x + 45}{15} = \frac{x + 15}{5}$.

Number of heads = $x + 2x + 45 + \frac{x + 15}{5} = \frac{16x + 240}{5}$.

Number of feet = $4x + 4 \times 2x + 2 \times 45 + 2\left(\frac{x + 15}{5}\right)$
 $= 4x + 8x + 90 + \left(\frac{2x + 30}{5}\right) = \frac{62x + 480}{5}$.

$\therefore \frac{62x + 480}{5} - \frac{16x + 240}{5} = 186 \Rightarrow 46x + 240 = 930$
 $\Rightarrow 46x = 690 \Rightarrow x = 15$.

Hence, number of caretakers = $\frac{x + 15}{5} = \frac{30}{5} = 6$.

430. Man's basic pay per hour = $\text{₹} \left(\frac{200}{40}\right) = \text{₹} 5$.

Overtime charges per hour = 125% of $\text{₹} 5 = \text{₹} 6.25$.

Suppose the man worked overtime for x hours.

Then, $200 + 6.25x = 300 \Rightarrow 6.25x = 100 \Rightarrow x = 16$.

$\therefore$ Total number of working hours = $(40 + 16) = 56$.

431. Suppose the man works overtime for x hours.

Now, working hours in 4 weeks = $(5 \times 8 \times 4) = 160$.

$\therefore 160 \times 24 + x \times 32 = 4320 \Rightarrow 32x = 480 \Rightarrow x = 15$.

Hence, total hours of work = $(160 + 15) = 175$.

432. Suppose their paths cross after x minutes.

Then, $11 + 57x = 51 - 63x \Leftrightarrow 120x = 40 \Leftrightarrow x = \frac{1}{3}$.

Number of floors covered by David in

$\frac{1}{3} \text{ min} = \left(\frac{1}{3} \times 57\right) = 19$.

So, their paths cross at $(11 + 19)$ i.e., 30th floor.

433. Let the total number of steps be n .

Then, $\frac{30}{n - 26} = \frac{18}{n - 34} \Leftrightarrow 30(n - 34) = 18(n - 26)$

$\Leftrightarrow 30n - 18n = 552 \Leftrightarrow 12n = 552 \Leftrightarrow n = 46$.

434. Let the price of the uniform be $\text{₹} x$.

Then, servant's pay per day = $\text{₹} \left(\frac{200 + x}{10}\right)$.

$\therefore 5\left(\frac{200 + x}{10}\right) = 20 + x \Leftrightarrow 200 + x = 40 + 2x$

$\Leftrightarrow x = 160$.

435. Suppose he remained absent for x days.

Then, number of days on which he was present = $(20 - x)$.

$\therefore 60(20 - x) - 5x = 745 \Leftrightarrow 65x = 455 \Rightarrow x = 7$.

436. Let the fixed charge be $\text{₹} x$ and variable charge be $\text{₹} y$ per km.

Then, $x + 16y = 156$... (i) and $x + 24y = 204$... (ii)

Subtracting (i) from (ii), we get: $8y = 48$ or $y = 6$.

Putting $y = 6$ in (i), we get: $x = 60$.

$\therefore$ Cost of travelling 30 km = $\text{₹} (60 + 30 \times 6) = \text{₹} 240$.

437. Original contribution = $\text{₹} \left(\frac{D}{M}\right)$.

New contribution = $\text{₹} \left(\frac{D}{M - 3}\right)$.

$\therefore$ Required contribution = $\text{₹} \left[\frac{D}{(M - 3)} - \frac{D}{M} \right]$
 $= \text{₹} \frac{DM - D(M - 3)}{M(M - 3)}$
 $= \text{₹} \left(\frac{3D}{M^2 - 3M} \right)$.

438. $N \times 50 = (325000 - 300000) = 25000 \Rightarrow N = 500$.

439. Let number of boys = x . Then, number of girls = $(60 - x)$.

$\therefore x(60 - x) + (60 - x)x = 1600$

$\Leftrightarrow 2x^2 - 120x + 1600 = 0$

$\Leftrightarrow x^2 - 60x + 800 = 0$

$\Leftrightarrow (x - 40)(x - 20) = 0$

$\Leftrightarrow x = 40$ or $x = 20$.

So, we are not definite, Hence, data is inadequate.

440. Original share of 1 person = $\frac{1}{8}$. New share = $\frac{1}{7}$.

Increase = $\left(\frac{1}{7} - \frac{1}{8}\right) = \frac{1}{56}$.

$\therefore$ Required fraction = $\frac{(1/56)}{(1/8)} = \left(\frac{1}{56} \times 8\right) = \frac{1}{7}$.